



EnerTraffic Radar-Vision Fusion Sensor Training

Location: Shenzhen, China

Website: <https://enertraffic.com>



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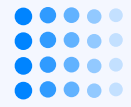
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Challenges & Evolution of Transportation Management



Challenges of Transportation Management



Single-Sensor Dependency

Traditional systems are often limited to single sensors, such as video, resulting in incomplete traffic data collection and affecting decision-making accuracy.

Data Silos

Data from different sensors is difficult to integrate, creating information silos that hinder comprehensive analysis and impede the precise optimization of traffic management.

Adverse Weather Failure

In adverse weather conditions such as rain, fog, and haze, the effectiveness of traditional video surveillance is significantly diminished, posing major challenges for traffic management.

Complex Event Missed Detection

Traditional systems have a high missed detection rate for complex events such as accidents or congestion, leading to delayed responses and impacting traffic flow efficiency.

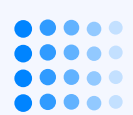
Inaccurate Traffic Parameters

Due to the inherent limitations of the sensors, the accuracy of acquired traffic parameters—such as vehicle speed and flow—is insufficient, making it difficult to provide reliable data support.

Poor Tracking & Identity Continuity

In scenarios involving occlusions or dense traffic flow, targets are easily lost, IDs switch frequently, and trajectories become fragmented.





Intelligent Transportation Architecture



Application Layer

Traffic Signal
Intelligent
Scheduling

Intersection
Event
Monitoring

Emergency
Incident
Alerts

Vehicular
Network Data
Services

Traffic
Efficiency
Evaluation

Services and
Applications

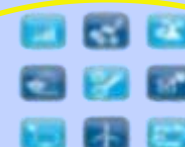
Analysis Layer



Map Engine



Data Engine



Open Platform

Data Fusion
and Mining

Cloud Platform

Cloud
Computing



Data Center



Data Computing
and Storage

Network Layer

Communica
tion Net

IOT

Internet

Data
Transmission

Edge
Computing &
AI Layer

Control (Single-Point or Area Traffic
Scheduling)

Perception (Multi-Data Fusion Computing,
Driving Data AI Computing)

Edge
Computing

Sensing Layer



Camera



Radar



Camera&Radar
Fusion Device



RFID

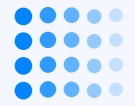


Environmental
Sensors



Beidou&GPS

Data
Collection



Evolution of the Sensing Layer

Intelligentization

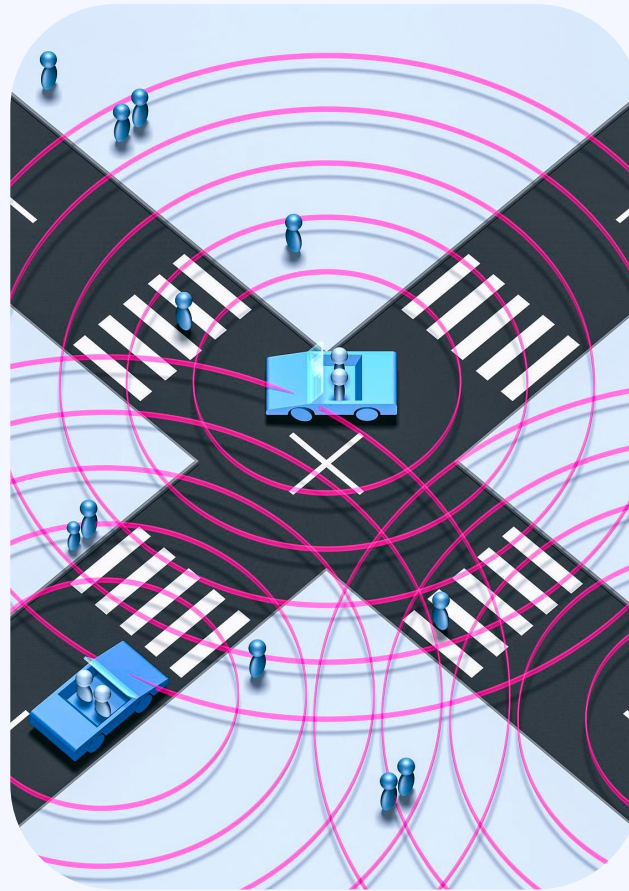
Modern traffic management must achieve an intelligent upgrade from "visibility" to "comprehensibility, accuracy, and all-weather capability" in order to tackle increasingly complex traffic environments.

Integration

From "Single Sensing" to "True Spatiotemporal Fusion." By deeply integrating sensors based on different physical principles at the hardware, data, and algorithmic levels, perception results superior to the simple combination of individual sensors are achieved.

Coordination

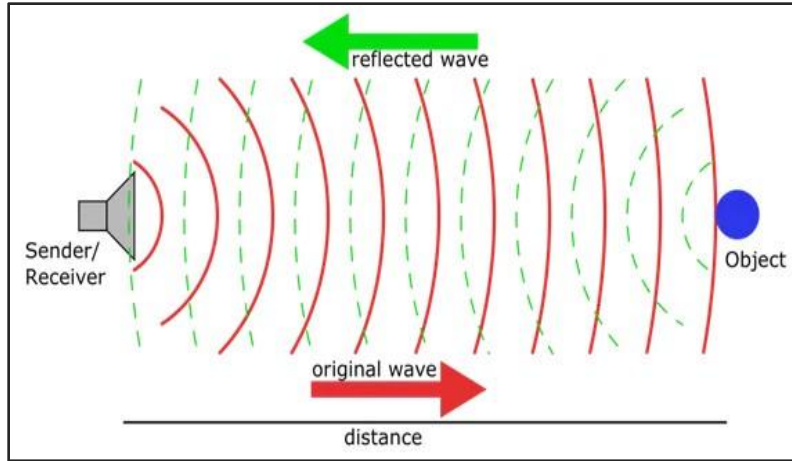
From "Isolated Point Detection" to "Road & Network Collaborative Sensing." This enables perception devices deployed at different locations to "communicate," achieving seamless and stable tracking of a target (especially vehicles) across continuous spaces through spatiotemporal synchronization and trajectory relay.



02

Radar-Video Fusion Technology Brief

Technical Principles of Millimeter-Wave Radar



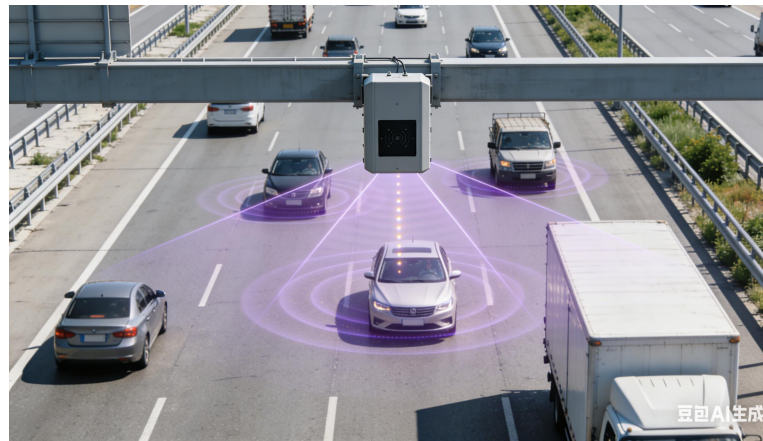
Technical Advantages

- ✓ All-weather, all-hours reliable operation.
- ✓ Direct measurement of speed & distance.
- ✓ Fast response with high real-time performance.
- ✓ Long detection range and wide coverage.



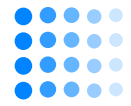
Working Mechanism

Millimeter-wave radar detects objects by emitting high-frequency electromagnetic waves and receiving the echoes reflected from them. By calculating the time difference between transmission and reception, it can precisely determine the target's distance. Utilizing the Doppler effect, it can calculate the target's speed. Additionally, through antenna arrays and algorithms, it can estimate the target's horizontal azimuth.



Disadvantages and Limitations

- Cannot "see" fine details, lacks information such as color and texture.
- Faces challenges in detecting low-speed/stationary targets.
- Relatively low spatial resolution.
- Unable to perceive semantic information of static environments.



Technical Principles of Video Surveillance

Working Mechanism

Optical Imaging

It utilizes optical lenses to capture images, converting light signals into electrical signals to form video footage.

Image Processing

It processes images through noise reduction, enhancement, compression, and other techniques to improve image quality, preparing them for subsequent analysis.

AI Recognition

Leveraging AI algorithms, it achieves object recognition, feature extraction, behavior understanding, and environmental interpretation.

Technical Advantages

- ✓ Rich information and strong semantic understanding, It provides the most intuitive and rich visual data.
- ✓ High spatial resolution and detail recognition capability.
- ✓ High static environment perception capability.
- ✓ Aligns with human intuition and is easy to review.

Disadvantages and Limitations

- Severely constrained by environmental and lighting conditions. Image quality drastically declines under low light, backlighting, heavy rain, dense fog, haze, strong glare, and similar adverse conditions.
- Difficulty in directly and accurately measuring physical quantities.
- Complex data processing and high computational load. relying on substantial backend AI computing power and incurring significant network costs.



Video and radar technologies complement each other.

Complementary Operating Conditions

All-weather reliability ensures continuous, uninterrupted 7x24 sensing capability.

Complementary Detection Information

Physical Quantities vs. Semantic Information, comprehensive perception of target information.

Complementary Target Resolution Capabilities

Point Cloud Tracking vs. Pixel Recognition, enabling comprehensive monitoring from macroscopic traffic flow to microscopic individuals.



Complementary Data Characteristics

Precision vs. Richness: Radar data provides precise Regions of Interest and spatial cues to reduce computational waste.

Complementary Environmental Perception

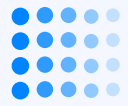
Dynamic Targets vs. Static Scenes: Associating the precise trajectories of dynamic vehicles with static elements.

Complementary Functionality and Reliability

Cross-validation reduces the false alarm rate by an order of magnitude.

03

Main Functionalities of Smart Radar



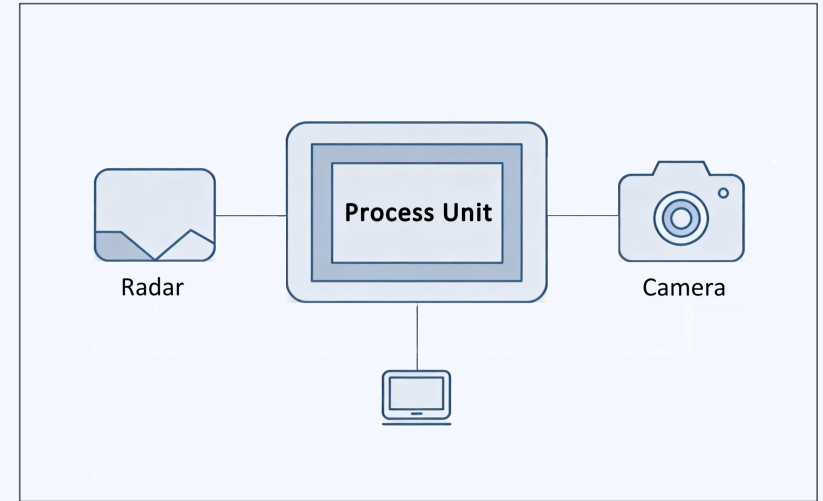
Hardware Platform



Camera

High-definition CMOS sensor for clear images capture, supporting vehicle type recognition, LPR, pedestrian detection etc., providing rich semantic infor for traffic monitoring. 3 cameras included:

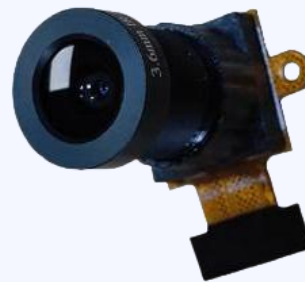
- 5MP for traffic statistics
- 5MP for license plate recognition
- Night vision camera



Radar

High-frequency millimeter-wave radar for long-range, high-precision target detection. Accurate vehicle positions and speeds, around the clock operation, reliable data support for traffic management.

- 200+ target tracking
- maximum of 400m

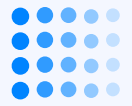


Processing unit

Powered by a high-performance processor, featuring an innovative multi-heterogeneous unit acceleration architecture.

- 2 RISK-V core
- Built-in new-generation KPU intelligent AI computing unit





Functionalities of Smart Radar



Multi-dimensional data output

The real-time output of the targets position, orientation, speed, size, type and confidence level, etc. provides comprehensive and accurate information for traffic management.

All-weather and all-time perception

Stable and reliable perception ability even in harsh environment.

Data fusion and visualization

The backend interface provides an intuitive display, perfectly overlaying radar trajectories with video images for clear visibility, and offers accurate and intuitive support for traffic management.

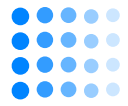


Reliable target recognition and classification

Visual AI technology used to perform concretized recognition on radar-detected targets and accurately differentiates among cars, trucks, pedestrians, cyclists, obstacles, animals, etc.

High precision target tracking

Precise distance/velocity data from radar fused with feature data from cameras to generate continuous, stable and smooth motion trajectory for the targets.



Comprehensive Traffic Data Acquisition



Lane-by-Lane Traffic Flow Statistics

Accurate real-time traffic flow data from each lane



Average Speed

Average speed of vehicles monitored in real time to reflect the smoothness of traffic



Space Occupancy Rate

Quantifies the proportion of road space occupied by vehicles in a specific period of time



Time Headway and Space Headway

Time interval and actual distance between adjacent vehicles, providing data support for safe driving analysis



Queue Length

Accurate queue length to timely grasp intersection congestion status



LPR

Accurately identify motor vehicle license plates



Vehicle Type

Refined vehicle classification, Such as small vehicles, large vehicles, motorcycles/non-motorized vehicles, etc.



Traffic Violations

Detecting traffic violations, such as red-light running, etc.



Traffic Accident

Traffic accidents detection



All-Weather and All-Time Perception



Cability of mm-Wave radar to penetrate rain and fog

mm-Wave radar can penetrate through severe weather conditions like heavy rain and dense fog, tracking vehicle trajectories with high accuracy even in visibility below 100 meters, ensuring stable output of core data.



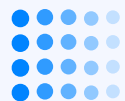
Low illumination and night tracking capability

HD night vision camera can still capture clear vehicle images in no-illumination nighttime environments
mm-Wave radar can detect targets hundreds of meters away without light



Adaptive Design for Severe Weather

The device features IP67-rated waterproof and dustproof enclosure, and maintains stable operation across temperature range of -40°C to +85°C and under severe conditions including rain, snow, and fog



Data Fusion and Visualization



01

Radar-Video Fusion

Radar data and video images are perfectly fused and complement each other, improving the accuracy of target detection and tracking

02

Intuitive Info Display

Fused data presented in intuitive info boxes and dynamic graphs, enabling quick understanding and decisions

03

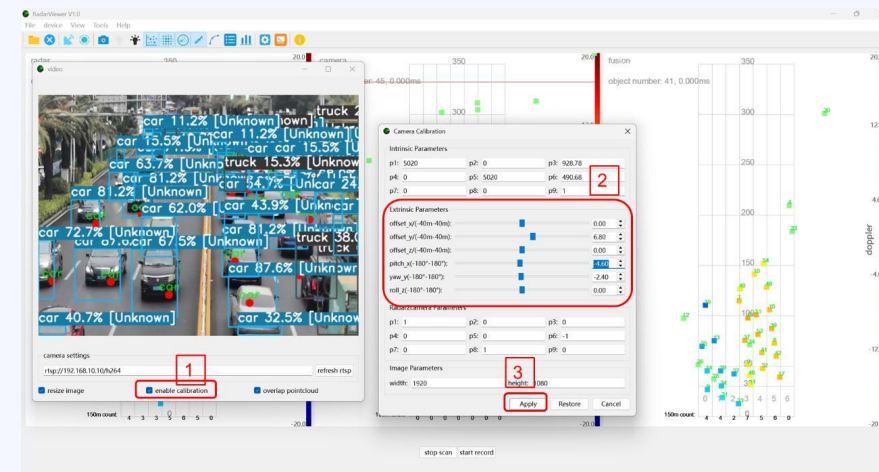
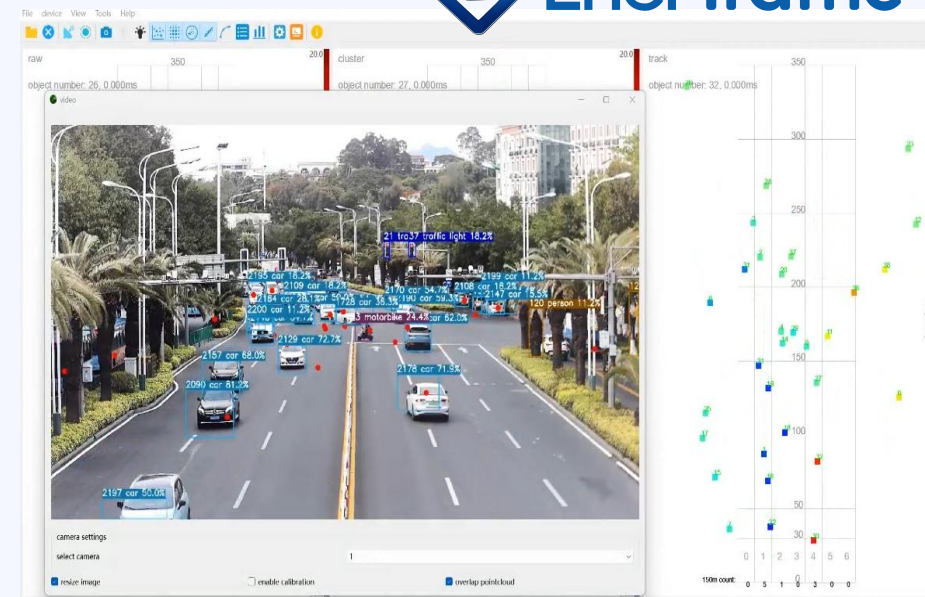
Dynamic Effect Enhancement

Highlight target trends and density changes using visualization techniques like animations

04

Convenient Operation

Supports simple visualized operations allowing users to flexibly adjust view details and scope





Reliable Target/License Plate Recognition



Video and radar cooperate to perform fused recognition, enhancing efficiency and accuracy

Camera: recognition and classification of targets and license plate mainly relies on camera images and AI algorithms

Radar: provides precise "candidate regions", greatly improving AI efficiency and reliability

- Radar screening: radar first detects targets with distance, orientation, speed and size.
- Guide AI to Focus: The system maps the radar targets position and size to the camera's image, forming a precise region of interest (ROI). This eliminates the need for full-image search and analysis.

- Scenario Example 1: At night or in heavy fog, image quality is extremely poor, so AI may be unable to determine whether the dim outline is a car or a cargo container. However, the radar clearly detects that the target has a radial speed of 60 km/h, thus it can almost be concluded that this is a vehicle rather than a stationary object.
- Scenario Example 2: When a partially obscured vehicle (e.g., emerging from behind a barrier) is detected, the camera may only capture a corner view with low recognition confidence. However, the radar can detect the reflected waves from the obscured portion, providing complete target dimensions and motion status to assist the AI in making accurate judgments.

High Precision Target Tracking



Radar Data Acquisition and Feature Extraction

Radar can accurately measure the core parameters of the target such as distance, velocity and angle, and has strong anti-interference ability and is suitable for harsh environment. However, it cannot track the target according to its characteristics, and it is easy to misjudge the static target.



Video Data Acquisition and Target Detection

AI vision technology acquires target details like orientation, vehicle type, and license plate, offering enhanced intuitiveness to assist in target classification. However, it is significantly affected by weather conditions such as rain and fog, and its ranging accuracy is not high.

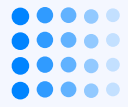


Multi-Sensor Data Fusion Algorithm

The algorithm is used to fuse radar and video data, achieving the complementation of information such as distance, speed, angle, and vehicle type. Thereby, it realizes accurate matching and real-time update of target trajectories, and improves the comprehensive information of targets.

04

Application Scenarios



Application Scenarios



Event Detection and Alerts

Monitor traffic incidents in real time, such as stopped vehicles, wrong-way driving, or collisions, and automatically trigger alarms while linking with video for verification



Radar Triggering and Video Evidence Linkage

E.G speeding Scenario: Utilize radar speed measurement combined with video trajectory verification, as well as multi-zone coordinated speed detection, to identify vehicle speeding.



Ramp Metering and Merge Warning

Dynamically control ramp signals based on main road traffic flow to prevent merging conflicts. Monitor vehicle speed and spacing in merging areas and alert drivers via variable message signs.



Safety Assurance in Adverse Weather

Under adverse weather conditions, it can monitor traffic flow information such as vehicle speed and spacing, thereby issuing corresponding guidance information to prevent accidents from occurring.



Intelligent Traffic Light Adjustment

Monitor real-time information such as traffic flow and queue length in all directions, automatically adjust traffic light timing to reduce idle time and congestion, or ensure priority passage for special vehicles.



Traffic Flow Monitoring&Analysis

Accurately count road section traffic volume and related info.
Automatically identify peak hours to provide a basis for dynamic speed limits, tidal flow lanes, and other measures to support traffic optimization.



Intelligent Traffic Light Adjustment

Improve traffic efficiency and alleviate urban congestion

By optimizing signal timing, the system reduces ineffective waiting time for vehicles at intersections, allowing more vehicles to pass through per unit of time. This effectively "widens" the road without altering the physical space.

Enhance traffic safety and reduce accident risks

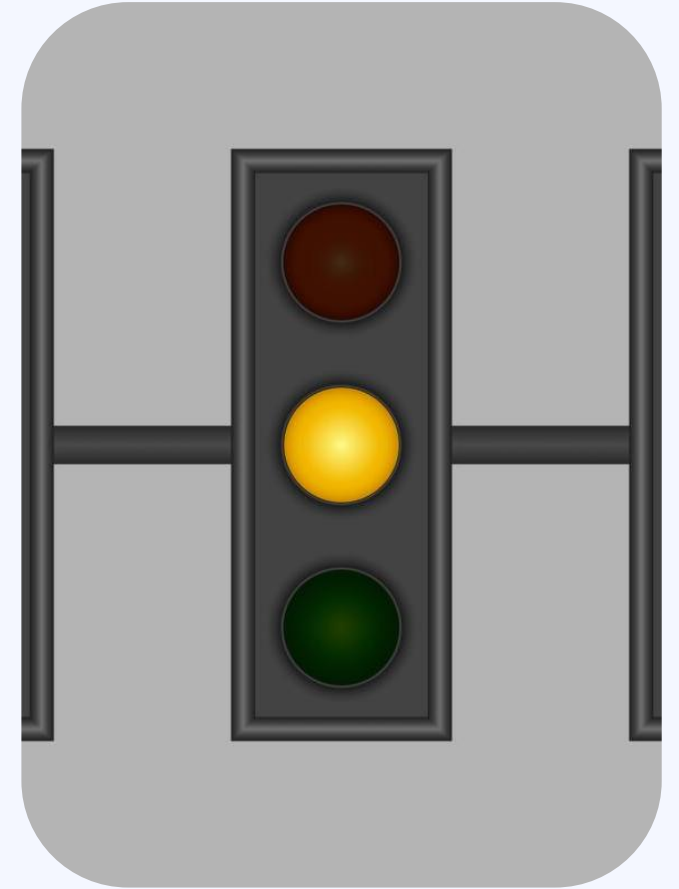
By optimizing signal phasing and sequencing, the system reduces the time and likelihood of conflicts between vehicles and pedestrians from different directions, while also decreasing drivers' tendency to rush through yellow lights due to waiting anxiety.

Achieve energy conservation and environmental protection

Frequent vehicle starts, stops, and idling are primary contributors to energy consumption and pollution. Intelligent regulation reduces parking frequency and idling time, directly lowering fuel consumption and carbon emissions by approximately 20%.

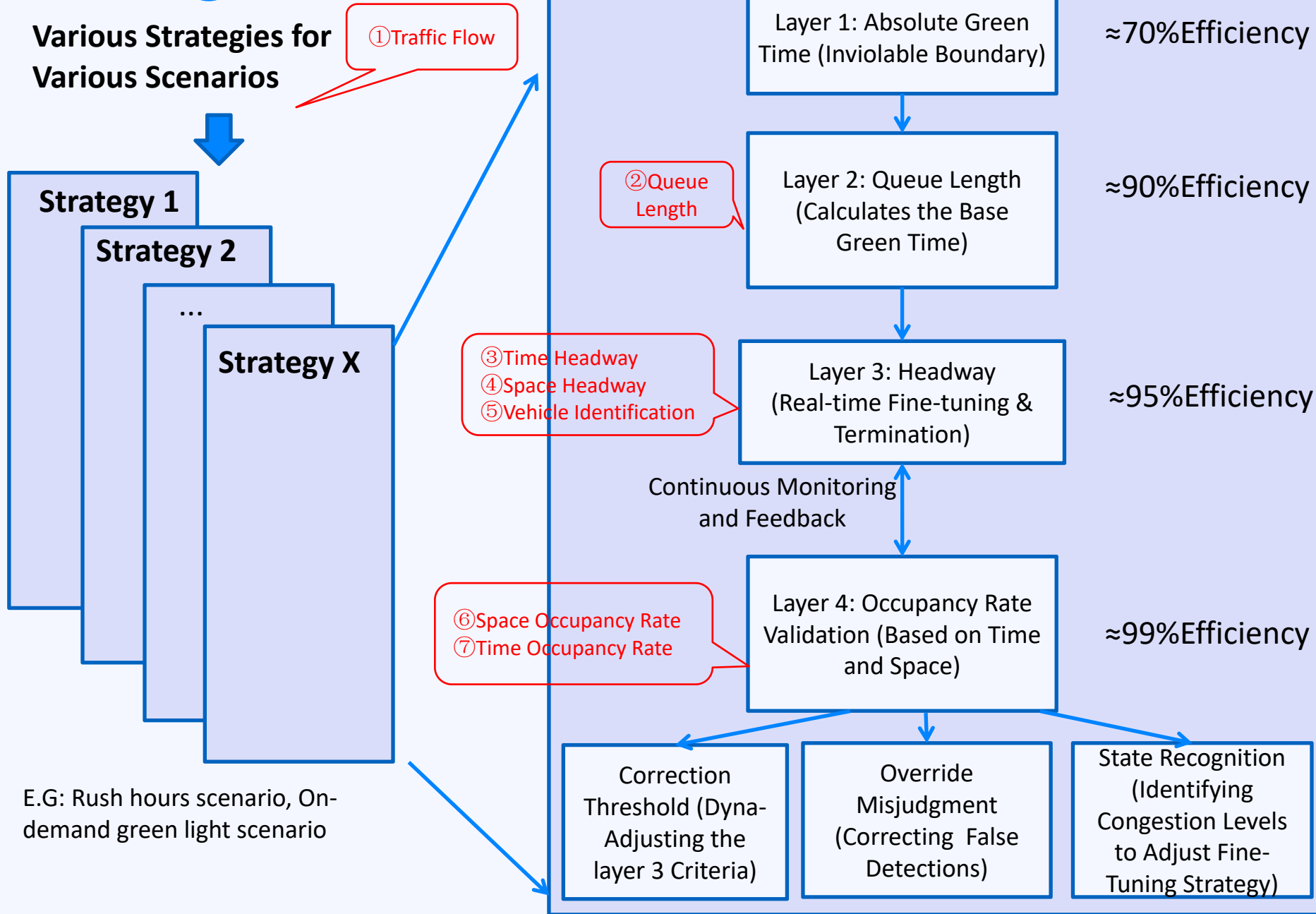
Generate direct and indirect economic value.

Saves time costs for all travelers (time is money), reduces fuel and maintenance expenses for vehicles, promotes commercial circulation and consumption, improves the utilization efficiency of existing infrastructure, and reduces capital investment in new construction.



Traffic Light Process

Various Strategies for Various Scenarios



Right-Turn Light Strategy:

- Base strategy coupled with straight through movement
- Four-layer strategy
- Coordination with opposing sensors

Left-Turn Light Strategy:

- Four-layer strategy
- Pedestrian volume detection

- Coordination of policies for different directions at an intersection
- Coordination between intersections

Event Detection and Alerts

Traffic Safety

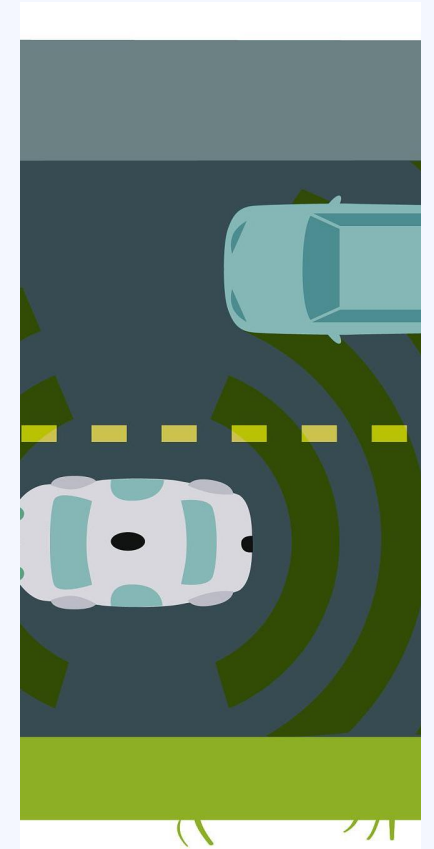
The radar-vision integrated system technology enables real-time detection of traffic accidents, sudden stops, wrong-way driving, and congestion, ensuring smooth traffic flow and public safety.

Traffic Violations

The radar-vision integrated system technology, through high-precision monitoring, effectively identifies traffic violations such as running red lights, not following lane guidance, crossing lines, lane changes, and speeding.

Traffic Order Management

Applied to traffic management, the radar-vision integrated system technology improves intersection overflow, reduces conflicts between motorized and non-motorized traffic, and provides pedestrian crossing safety warnings through precise perception and intelligent analysis.



Radar Triggering and Video Evidence Linkage



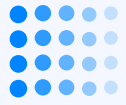
Dual Perception Verification

Radar often detects vehicle anomalies first, such as speeding, and immediately triggers high-definition cameras for video evidence collection, forming a dual-evidence chain of "radar alert + video verification."



Data Synchronization

Ensures time synchronization between radar speed measurement data and video footage, meeting judicial evidence requirements. Enables coordinated detection and evidence collection across multiple devices in the same area (e.g., for speed measurement).



Ramp Metering and Merge Warning



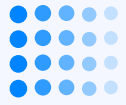
By using the radar-vision integrated system to monitor main road traffic density in real time, the timing plan for ramp signals is dynamically adjusted to effectively prevent congestion on the main road. When main road traffic approaches saturation, the ramp red light cycle is automatically extended to reduce the impact of merging vehicles on the main road.

Dynamic control of ramp signals based on main road traffic flow



vehicle spacing warnings in merging areas.

Based on the speed of the main road, the minimum safe merging distance can be calculated. When it is detected that the actual merging distance does not meet the safety threshold, a tiered warning is issued using LED flashing lights and audible-visual alarms.



Safety Assurance in Adverse Weather



01

Adaptability to Harsh Environments (Fog/Rain/Nighttime)

The radar-vision integrated system utilizes millimeter-wave radar and video fusion technology, maintaining an effective monitoring range of over 300 meters even under conditions of dense fog with very low visibility or during nighttime.



02

Dynamic Vehicle Monitoring Functions

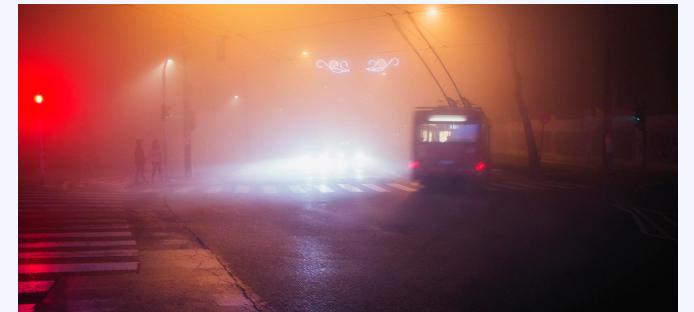
The system can capture real-time changes in vehicle speed and automatically triggers a warning mechanism when the distance between adjacent vehicles falls below the safety threshold.



03

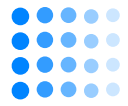
Intelligent Guidance Systems

Dynamically displays speed limits on variable message signs and implements tiered warnings using audible and visual alarm devices.



05

Deployment Guide



Preliminary Planning and Site Design



Demand Analysis and Scenario Adaptation

Based on traffic management objectives (such as signal optimization, incident detection, or non-contact enforcement), clarify the functional requirements for the radar-vision integrated system. Considering factors like the number of lanes at the intersection, traffic flow, and surrounding environment, determine the required perception range and accuracy for the equipment.

Equipment Site Selection Principles

Choose open, unobstructed locations to ensure radar and video coverage of all target lanes and key areas at the intersection. Consider installation height and angle to avoid direct strong light exposure and multipath interference, ensuring optimal radar detection and video imaging performance.

Installation Environment Assessment

Assess the stability of the power supply, network transmission conditions, load-bearing capacity of the pole, and protection requirements (such as IP rating, temperature and humidity adaptability) at the installation site. Plan cabling solutions in advance to lay the groundwork for the long-term stable operation of the equipment.



Equipment Mounting and Stability Requirements

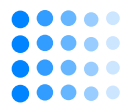
Use dedicated brackets and fasteners for installation to ensure the equipment is secure and vibration-free, with horizontal and vertical deviation not exceeding $\pm 2^\circ$. This prevents impact on radar speed measurement accuracy and video tracking stability due to equipment shaking.

Power and Network Connection Standards

Connect to a stable 12V power supply. Utilize a Gigabit Ethernet interface with network transmission latency $\leq 100\text{ms}$ to ensure real-time data upload. Cables must be waterproofed, lightning-protected, and comply with outdoor installation standards.

Protection and Environmental Adaptation Measures

The equipment housing must have a protection rating of at least IP67 and operate within a temperature range of -40°C to $+85^\circ\text{C}$. If necessary, install sunshades and cleaning devices on the lens to mitigate the impact of adverse weather conditions such as rain, fog, and sand.



Lane Configuration

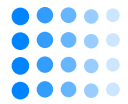
Use software to adjust and set lane width, center points , detection range configuration, and other parameters.

Radar-Video Spatial Synchronization

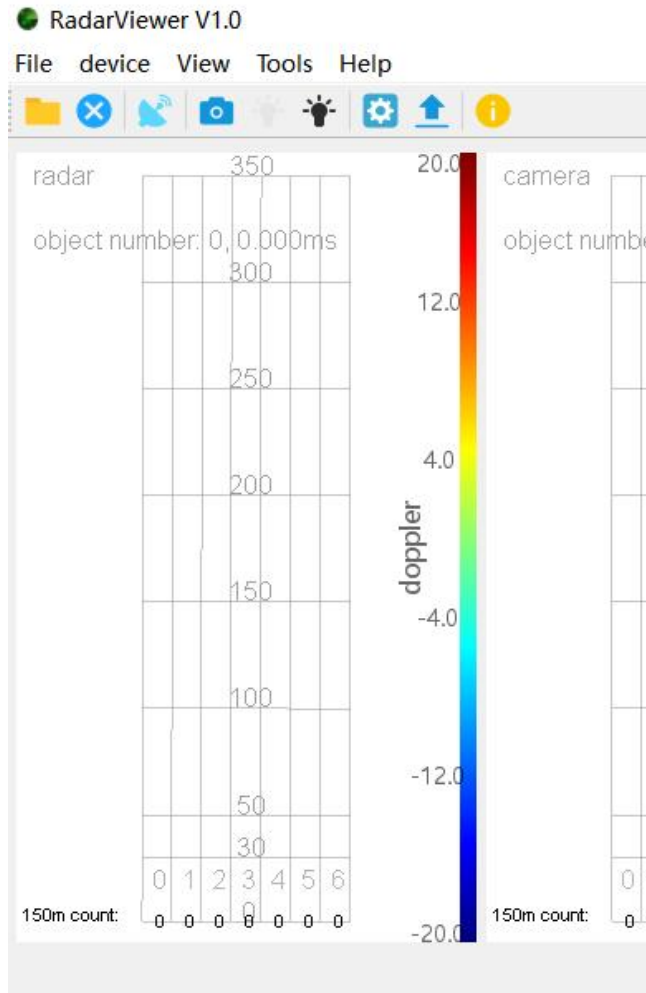
Within the fusion platform software, map and match the radar-detected target coordinates (distance and angle) with the pixel coordinates of the video image. By adjusting parameters, align the radar trajectory with the target position in the video, achieving real-time overlay of "point tracks and images."

Other Traffic Parameter Settings

Based on monitoring and management requirements, configure other relevant parameters, such as detection time, congestion level thresholds, etc.



Connect to Smart Radar



1. Insure the network is correct
2. Open the PC software , click Device → Open Device in the menu bar, or use the toolbar icon .

radar device

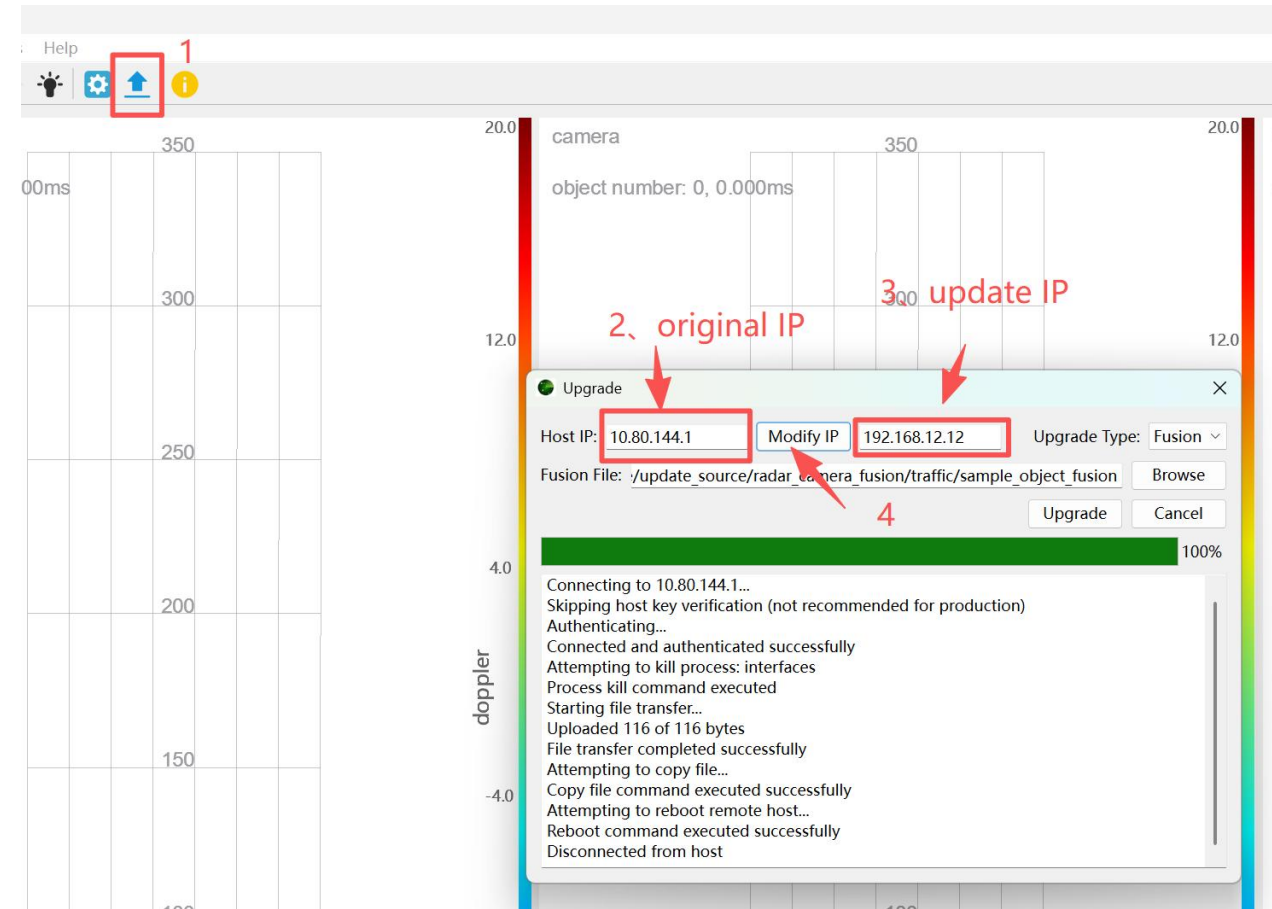
Board Setting

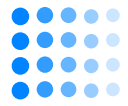
Board IP: Data Port:

3. Click “Apply” to establish the connection between the PC software and the Smart Radar.
4. Click View → Camera in the menu bar, or click the corresponding icon  in the toolb to open the camera window.

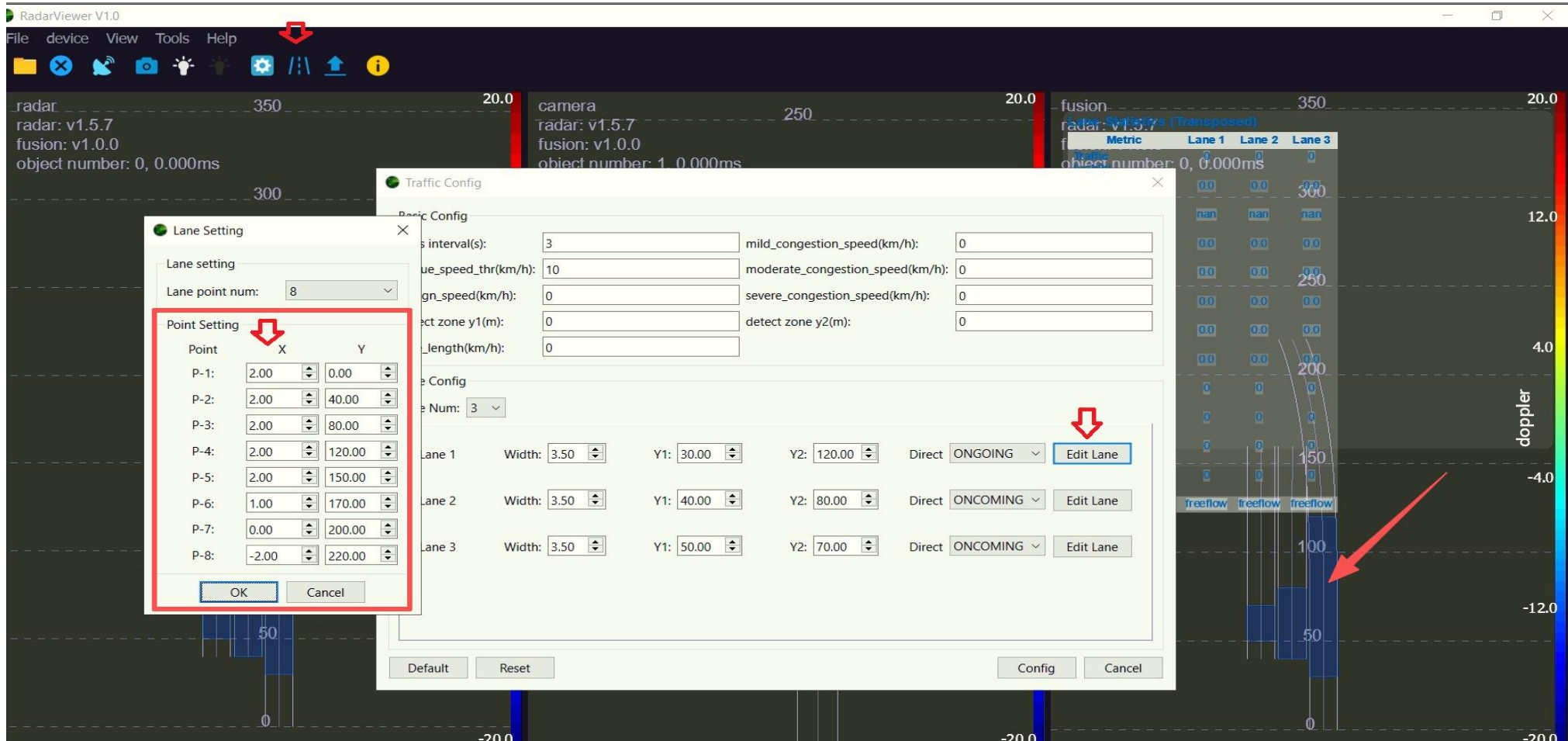
Modify the IP Address

1. Open the PC client software
2. Click **Device** → **Open Device** and connect to the device using its current IP address
3. Click the **Upgrade** button to enter the Upgrade interface
4. Enter the new IP address in the input box to the right of **Modify IP**
5. Click the **Modify IP** button
6. Wait for the device to restart, then connect to the device using the new IP address





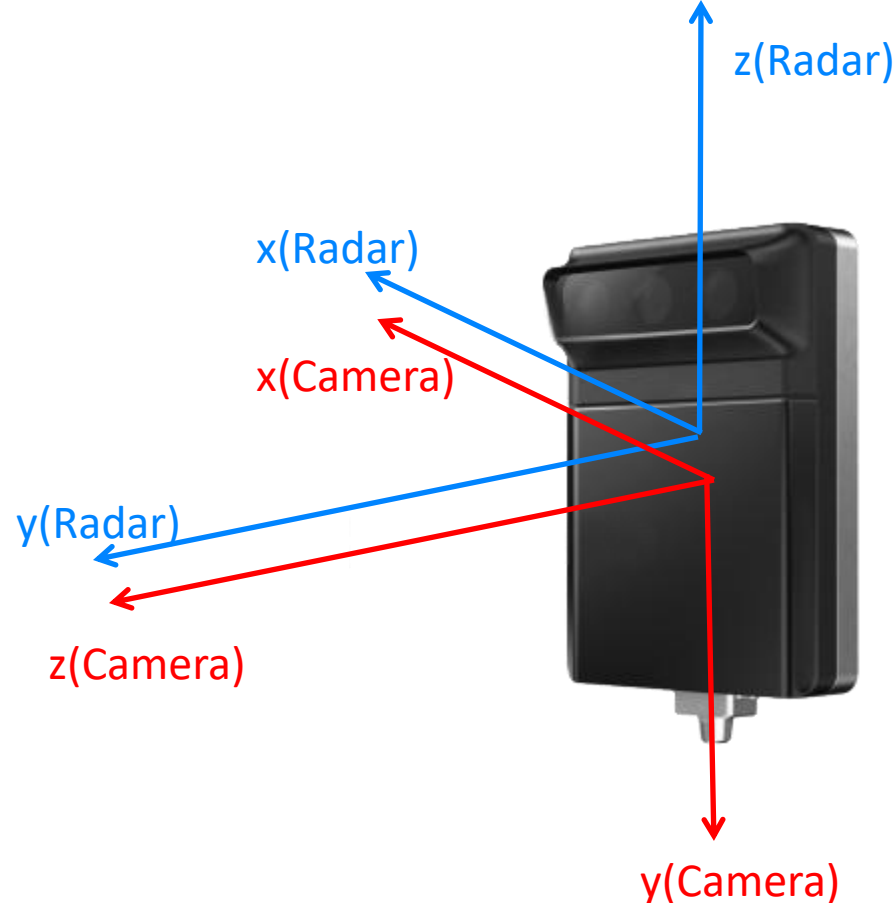
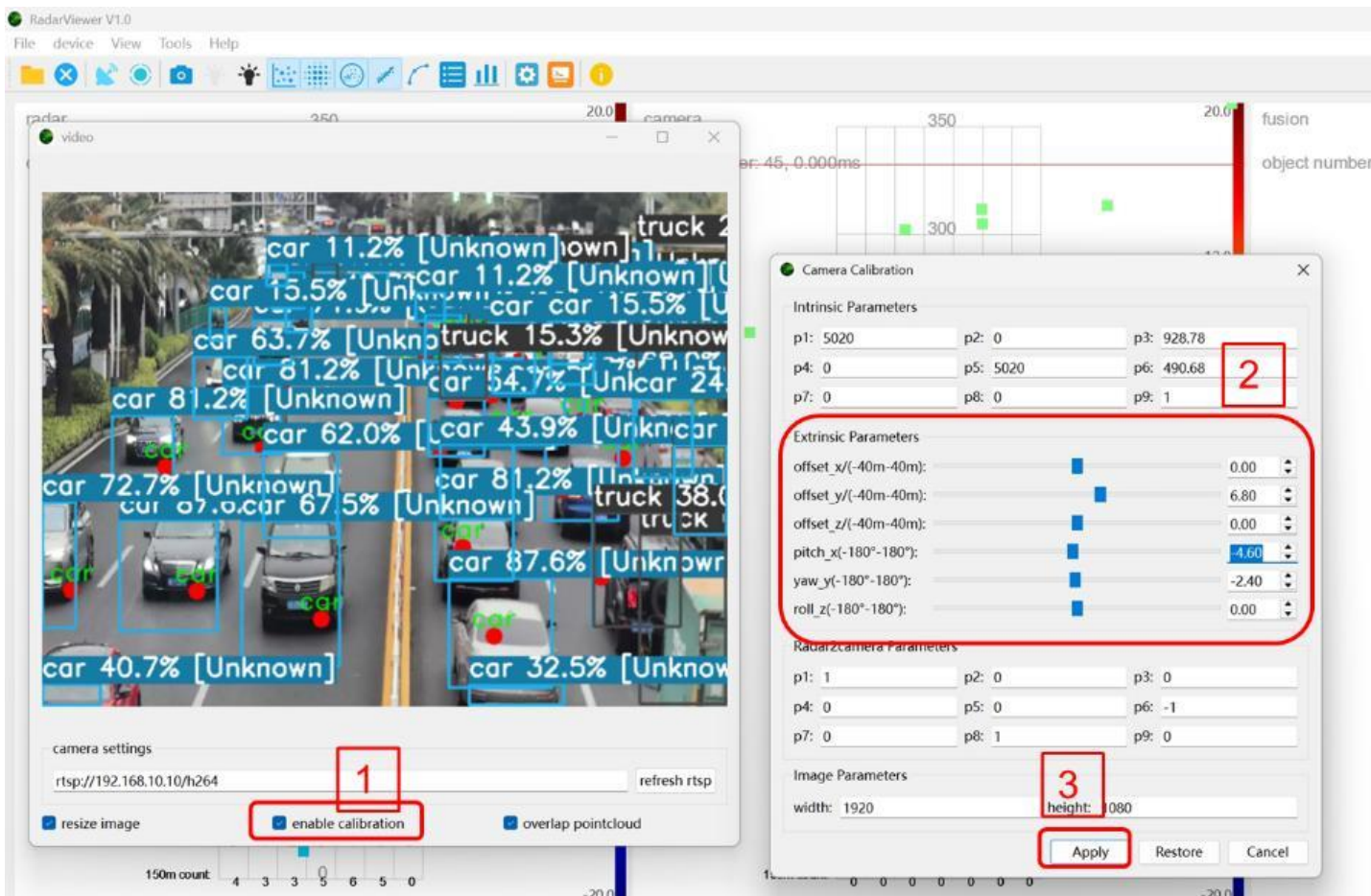
Lane & Traffic Configuration



Up to 16 lanes can be configured. Each lane can define up to 10 points as its centerline to simulate curved paths.



Sensor Calibration



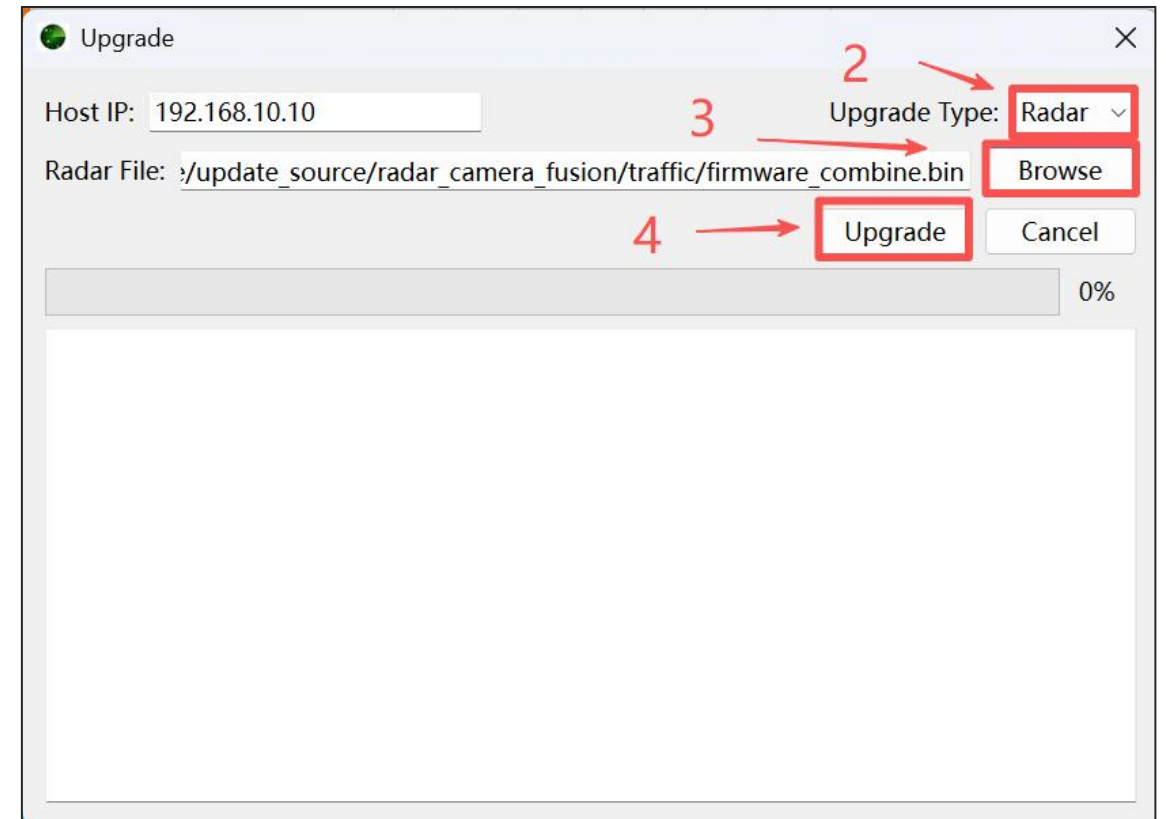
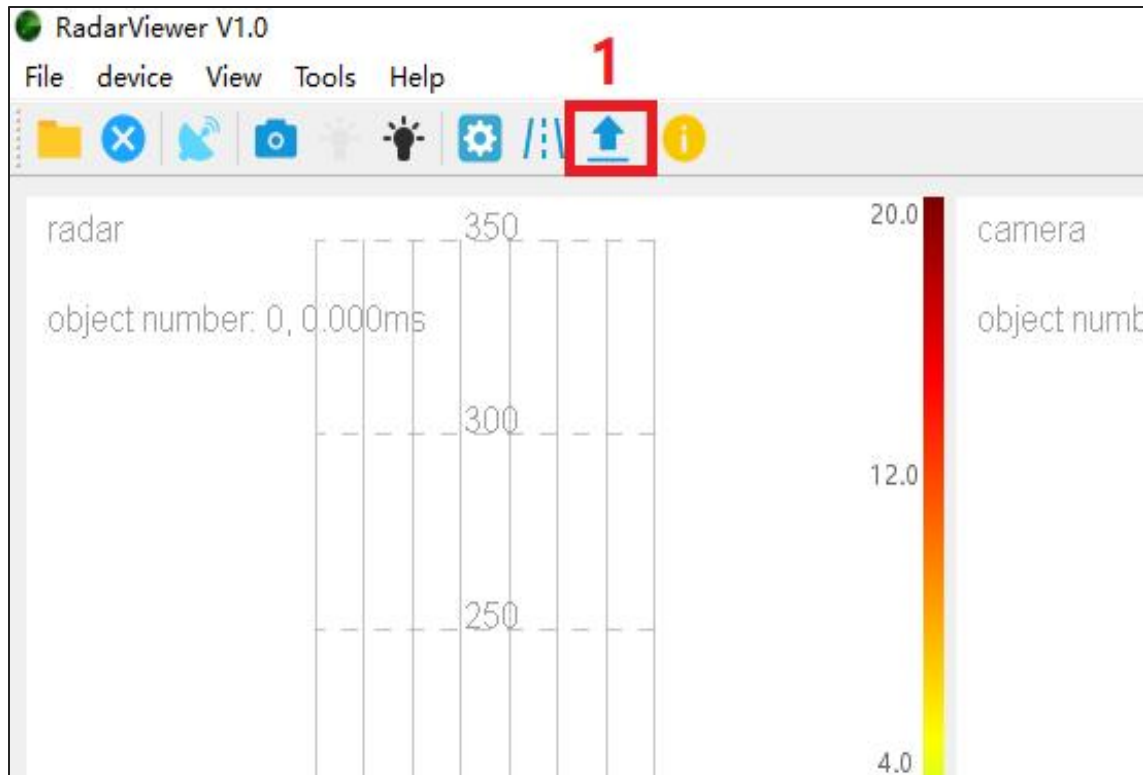
1. “Enable Calibration” checkbox is selected
2. adjust the radar point cloud position within the image by using the extrinsic parameter adjustment panel.
3. When the radar point cloud aligns precisely with the visual targets, calibration is considered complete

Notes: Recommended to adjust the 4 vparameters (offset_y , pitch_x, yaw_y ,roll_z)

Sensor Type	X-Axis Direction	Y-Axis Direction	Z-Axis Direction
Camera	Right	Down(towards ground)	Forward
Radar	Right	Forward	Up(towards sky)

Firmware OTA Upgrade

1. Click the “Upgrade” button  to enter the upgrade interface.
2. In the interface, set Upgrade Type to “Fusion” or “Radar” according to actual upgrade type .
3. Click the “Browse” button and select the fusion firmware file to be upgraded.
4. Click the “Upgrade” button to start the upgrade process.
5. Once the upgrade is complete, a success message will be displayed.



License Plate Recognitio

First, the license plate localization algorithm quickly and accurately identifies the license plate region within the image. Then, the license plate recognition algorithm performs character recognition on the localized region, outputting complete license plate number information.

Vehicle Attribute Recognition

It enables refined vehicle classification, including recognition of 10 vehicle body colors and 9 vehicle model categories.

Color	Type
yellow	sedan
orange	suv
green	van
gray	hatchback
red	mpv
blue	pickup
white	bus
golden	truck
brown	estate
black	



06

Specification & Parameters

Some internal technical details are not shown in this public document.
If you require this information for your project evaluation, please contact us
via our official website.

EnerTraffic
Official Website: <https://enertraffic.com>

Device Specifications

General Specifications

Product name	Smart Radar
Working voltage	9-36V
Working current	Average: 220mA / Peak: 310mA
Power consumption	<4.6W
Interface	Ethernet
Operational environment	-40°C ~ 85°C, 0 ~ 95% RH (no condensation)
Storage environment	-40°C ~ 105°C, 0 ~ 95% RH (no condensation)
Protection level	IP67

Device Specifications

Radar Dedicated Specification

Working frequency range		79~81GHz
Distance	Detection Range	350m
	Accuracy	±0.2m
Velocity	Scope	337km/h
	Accuracy	±0.05km/h
Azimuth para	Scope	14°
	Accuracy	±0.5°@0°
Pitch para	Scope	20°
Max number of detectable targets		200+
Scan frequency		≈10Hz
Working Temperature		-40°C to 85°C

Camera Dedicated Specification

	Camera1 (Traffic)	Camera2 (LPR)	Camera3 (Night)
Image resolution	5MP(2592×1944)	5MP(2592×1944)	2MP(1920 × 1080)
Sensor size	1/4 inch	1/4 inch	1/2.8 inch
Focal length	16mm	25mm	3.5mm
Maximum frame rate	30FPS	30FPS	30FPS



07

Maintenance Tips

Regular Maintenance Precautions



Equipment Cleaning and Protection

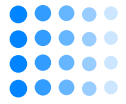
Regularly clean dust and stains from the equipment housing and lens surface to avoid obstruction that could affect sensing accuracy. Check the sealing integrity of the protective housing to ensure the IP protection rating meets standards, preventing water and dust from entering internal components.

Power and Network Stability Checks

Regularly inspect the power supply lines to ensure they are functioning properly and that the voltage remains stable within the equipment's operating range. Conduct periodic tests on the network connection status to guarantee smooth data transmission and avoid data loss or delays caused by network interruptions.

Environmental Adaptability Monitoring

Monitor changes in the operating temperature and humidity of the equipment environment to ensure it remains within the specified tolerance range. After extreme weather events (such as high heat, severe cold, or heavy rain), promptly check the operational status of the equipment to prevent malfunctions caused by environmental anomalies.



Common Troubleshooting

01

No Data Troubleshooting

1. Check the device power supply is normal.
2. Troubleshoot the network link by verifying whether Ethernet cable ports are loose and if the switch is functioning properly.
3. If both power and network connections are intact, inspect the device's internal core modules for proper operation.

02

Trajectory Drift Troubleshooting

Check whether the spatial synchronization calibration between radar and video is accurate, and re-perform radar-vision fusion calibration. Inspect whether the equipment installation is secure and if any vibration or external force has caused positional deviation. If the issue persists, review the radar module parameter settings for appropriateness and perform parameter recalibration if necessary.

03

AI Recognition Issues

Possible causes include a dirty or contaminated lens, light pollution issues, or physical displacement of the device. Therefore, it is necessary to clean the lens, remove sources of light pollution, and correct any physical misalignment.

Public-release version.

Some technical details are omitted in this public document.

If you need further information for evaluation, please feel free to contact us.

EnerTraffic

Official Website: <https://enertraffic.com>

Submit your inquiry via the contact form on our website.

The background is a solid light blue color. There are four decorative circles of varying sizes and shades of blue. One small circle is in the top-left corner. A large circle is partially visible on the left edge. Another small circle is in the bottom-right corner. A fourth circle is partially visible on the right edge.

THANKS